

```

19. # We can't print to the last column on Windows without it adding a
20. # newline automatically, so reduce the width by one:
21. WIDTH -= 1
22.
23. print('Digital Stream, by Al Sweigart al@inventwithpython.com')
24. print('Press Ctrl-C to quit.')
25. time.sleep(2)
26.
27. try:
28.     # For each column, when the counter is 0, no stream is shown.
29.     # Otherwise, it acts as a counter for how many times a 1 or 0
30.     # should be displayed in that column.
31.     columns = [0] * WIDTH
32.     while True:
33.         # Set up the counter for each column:
34.         for i in range(WIDTH):
35.             if columns[i] == 0:
36.                 if random.random() <= DENSITY:
37.                     # Restart a stream on this column.
38.                     columns[i] = random.randint(MIN_STREAM_LENGTH,
39.                                                  MAX_STREAM_LENGTH)
40.
41.             # Display an empty space or a 1/0 character.
42.             if columns[i] > 0:
43.                 print(random.choice(STREAM_CHARS), end='')
44.                 columns[i] -= 1
45.             else:
46.                 print(' ', end='')
47.         print() # Print a newline at the end of the row of columns.
48.         sys.stdout.flush() # Make sure text appears on the screen.
49.         time.sleep(PAUSE)
50. except KeyboardInterrupt:
51.     sys.exit() # When Ctrl-C is pressed, end the program.

```

After entering the source code and running it a few times, try making experimental changes to it. The comments marked with (!) have suggestions for small changes you can make. On your own, you can also try to figure out how to do the following:

- Include characters besides just 1s and 0s.
- Include shapes besides lines, including rectangles, triangles, and diamonds.

Exploring the Program

Try to find the answers to the following questions. Experiment with some modifications to the code and rerun the program to see what effect the changes have.

1. What happens if you change `print(' ', end='')` on line 46 to `print('.', end='')`?